An Intelligent AI-Driven Decision Support System for Personalized Income Tax Compliance and Advisory in Sri Lanka

R26-DS-004



Project Proposal Report

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March 2026

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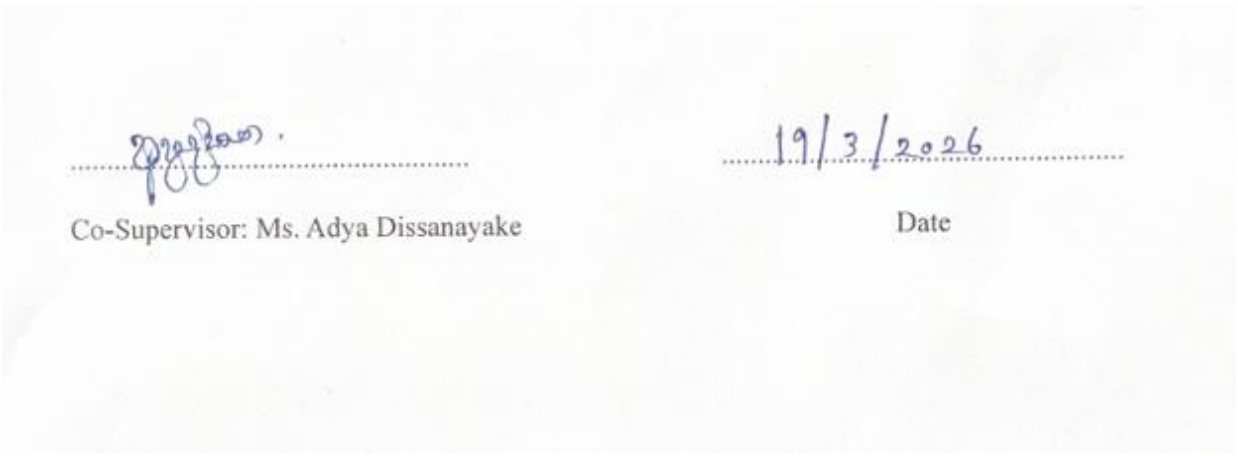
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Income tax calculation and planning can be complex for many individuals due to changing regulations, multiple income categories, and various deduction schemes. In Sri Lanka, most taxpayers rely on auditors or tax consultants to understand tax obligations and identify potential tax-saving strategies. This process can be time-consuming, expensive, and inaccessible for individuals who lack professional financial guidance. Although several digital tax systems exist globally, most of them focus mainly on tax calculation and filing rather than providing intelligent tax strategy recommendations tailored to Sri Lankan tax regulations. In addition, many existing AI-based financial advisory systems lack transparency, making it difficult for users to understand how recommendations are generated.

To address these limitations, this research proposes an **AI-driven Tax Strategy Optimization and Explainable Decision Support System** designed specifically for Sri Lankan taxpayers. The system analyzes financial data such as income sources, deductions, and applicable tax rules, and generates optimized tax planning strategies while ensuring compliance with the Inland Revenue Act of Sri Lanka. Machine learning techniques, rule-based tax reasoning, and explainable AI methods are integrated to provide personalized and transparent recommendations.

The proposed system aims to improve tax planning efficiency, reduce reliance on manual advisory services, and help users better understand their financial decisions. By providing accessible and explainable tax recommendations, the system can support individuals, tax professionals, and financial advisors while contributing to improved financial awareness and digital innovation in Sri Lanka's taxation ecosystem.

Keywords: Artificial Intelligence, Tax Strategy Optimization, Explainable AI, Decision Support Systems, Sri Lankan Taxation

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
XAI	Explainable Artificial Intelligence
IRD	Inland Revenue Department
API	Application Programming Interface
NLP	Natural Language Processing
DB	Database
IT	Information Technology
GDP	Gross Domestic Product
SDG	Sustainable Development Goals

Table 1:List Of Abbreviations

INTRODUCTION

1.1 Background and Context

1.1.1 Real-World Problem

Income tax calculation and tax planning are important financial activities for individuals and businesses in every country. In Sri Lanka, income tax is regulated by the **Inland Revenue Act No. 24 of 2017 and its subsequent amendments**, which define how taxes are calculated based on different income sources such as employment income, business income, investment income, and other income categories [1]. Although the legal framework clearly defines taxation rules, calculating tax liabilities and identifying optimal tax strategies can be complex due to frequent policy changes, multiple deductions, and various tax relief schemes.

In practice, many taxpayers rely heavily on **auditors, tax consultants, or accountants** to calculate taxes and provide advice on tax planning. This manual approach can be time-consuming, expensive, and sometimes inconsistent depending on the experience of the advisor. Individuals who lack professional financial knowledge may struggle to understand tax regulations and may either overpay taxes or unintentionally violate tax rules. As tax systems become more complex, the need for intelligent and automated decision-support tools becomes increasingly important.

Recent developments in **artificial intelligence (AI), machine learning (ML), and data analytics** have created opportunities to support financial decision-making processes through intelligent systems [2]. These technologies can analyze large volumes of financial data, identify patterns, and generate recommendations that help users make informed decisions. Applying such technologies to taxation systems could significantly improve efficiency, transparency, and accessibility of tax advisory services.

1.1.2 Who is Affected and Why It is Important

Taxation directly affects **individual taxpayers, businesses, auditors, and government authorities**. Individuals must ensure they accurately calculate their tax liabilities and comply with legal regulations. However, many taxpayers lack sufficient knowledge about tax rules, deductions, and available relief options. As a result, they often depend on professional tax advisors for guidance, which may increase financial costs and reduce accessibility for lower-income individuals.

Businesses also face similar challenges when managing corporate taxes, particularly when dealing with complex financial transactions or multiple income streams. From a government perspective, efficient tax administration is essential for maintaining economic stability and ensuring fair revenue collection. According to global studies, improved digital tax systems can significantly enhance compliance rates and reduce tax evasion [3].

Therefore, developing intelligent systems that assist taxpayers in understanding tax regulations and optimizing tax strategies can benefit both individuals and public institutions by improving transparency, efficiency, and financial planning.

1.1.3 Statistical Evidence (Local and Global Context)

Globally, governments collect trillions of dollars in tax revenue each year, making taxation one of the most critical financial mechanisms supporting national economies. According to the **Organisation for Economic Co-operation and Development (OECD)**, tax revenue accounts for approximately **34% of GDP in developed economies** [4]. Efficient tax systems therefore play a crucial role in economic development and public service funding.

In Sri Lanka, tax revenue represents a major portion of government income, with income tax contributing significantly to national fiscal resources. Reports from the **Department of Inland Revenue Sri Lanka** highlight increasing efforts to digitize tax processes through online services and electronic submissions [1]. However, despite these improvements, tax advisory and planning processes are still largely dependent on manual expertise from auditors and tax consultants.

Recent studies also suggest that AI-driven financial advisory systems can improve decision-making accuracy by analyzing complex financial datasets and recommending optimal strategies [5]. This demonstrates the potential of intelligent systems to support taxpayers in managing their financial obligations more effectively.

1.1.4 Analysis of Existing Solutions and Their Limitations

Several digital tax systems and financial software solutions have been developed globally to simplify tax filing and compliance. Popular tax software platforms such as **TurboTax, TaxAct, and H&R Block** provide automated tax calculation features and guided tax filing processes

[6]. These systems help users calculate their tax obligations based on predefined rules and input financial information.

In addition, research studies have explored the use of **machine learning and data mining techniques** for tax fraud detection, tax compliance analysis, and financial forecasting [7]. These systems demonstrate how AI technologies can support taxation processes by identifying patterns in financial data.

However, most existing tax systems focus primarily on **tax calculation and reporting**, rather than providing intelligent tax planning or optimization strategies. They typically require users to manually enter financial data and do not analyze different financial scenarios to suggest the most beneficial tax strategies. Furthermore, many systems lack **explainability**, meaning users cannot clearly understand why a particular recommendation or calculation result was generated.

In developing countries such as Sri Lanka, these limitations are even more evident because tax advisory services are still largely **manual and expertise-driven**, limiting accessibility for many taxpayers.

1.1.5 Proposed Solution and System Components

To address these challenges, this research proposes an **intelligent tax strategy optimization system** designed to assist individuals in analyzing financial information and identifying optimal tax strategies while maintaining compliance with Sri Lankan tax regulations.

The proposed system integrates **machine learning techniques, rule-based tax reasoning, and explainable AI models** to analyze user financial data and generate tax strategy recommendations. The system will process various income categories such as employment income, business income, and investment income, apply tax rules defined in the Inland Revenue Act, and evaluate different tax scenarios to identify strategies that can reduce tax liability legally.

The system architecture includes several components such as a **financial data processing module, tax rule engine, tax strategy optimization engine, and explanation module**. These components work together to analyze financial data, apply legal tax rules, and generate clear recommendations that help users understand the reasoning behind suggested tax strategies.

1.1.6 Relevant Sustainable Development Goal (SDG) Alignment

This research aligns with **Sustainable Development Goal (SDG) 9 – Industry, Innovation, and Infrastructure**, which promotes technological innovation and digital transformation to improve economic systems and infrastructure [8]. By developing an intelligent tax advisory system, this research contributes to digital innovation in financial governance and supports the modernization of taxation processes.

Furthermore, the system can also support **SDG 16 – Peace, Justice, and Strong Institutions**, which emphasizes transparent and accountable institutions. By improving accessibility and transparency in tax advisory processes, intelligent systems can enhance trust in financial governance and encourage better tax compliance.

1.1.7 Research Cluster Alignment

This research falls within the **Artificial Intelligence and Data Science research cluster**, as it involves the application of machine learning, optimization algorithms, and explainable AI techniques to support decision-making in financial systems. The project also intersects with the **Financial Technology (FinTech)** domain, where digital technologies are used to improve financial services and financial management processes.

By combining AI-based decision systems with financial analytics, this research contributes to the growing field of intelligent financial advisory systems that support individuals and organizations in managing complex financial decisions.

1.2 Literature Review

1.2.1 Digital tax services and existing tax systems

In recent years, taxation systems around the world have increasingly adopted digital technologies to improve efficiency and accessibility. In Sri Lanka, the Inland Revenue Department provides online services for tax registration, return submission, payment processing, and other administrative functions [1]. These digital platforms have made tax compliance more convenient for users and improved overall efficiency in tax administration. Similarly, global tax software systems such as TurboTax and other automated platforms help users calculate their tax liabilities and complete filing procedures more easily.

However, most of these systems are mainly designed for tax calculation and filing rather than providing intelligent support for tax planning. They require users to input financial data and then apply predefined rules to generate results. While this approach improves efficiency, it does not actively help users identify better tax-saving strategies or understand how different financial decisions impact their tax outcomes. This highlights a clear gap between simple digital tax processing and intelligent tax advisory systems.

1.2.2 AI in financial decision-making and advisory systems

Artificial Intelligence (AI) has been widely used in financial systems to support decision-making processes. According to Russell and Norvig, AI systems are capable of analyzing complex data, identifying patterns, and generating intelligent recommendations that assist users in making better decisions [2]. In financial domains, AI-based systems have been successfully applied in areas such as investment planning, risk analysis, and personalized financial advisory.

Research by Smith and Lee shows that AI-driven financial advisory systems can provide personalized recommendations based on user data and financial behavior [6]. These systems improve decision-making by analyzing multiple factors simultaneously and suggesting optimized solutions. Similarly, machine learning has also been applied in taxation for tasks such as auditing and policy analysis, where it helps identify patterns in financial data and improve efficiency [3].

Despite these advancements, most AI applications in taxation are focused on compliance, auditing, or prediction rather than tax strategy optimization. There is still limited research on systems that actively recommend personalized tax-saving strategies for individuals.

1.2.3 Optimization approaches in tax planning

Tax planning involves selecting the best financial strategies to minimize tax liability while staying compliant with legal regulations. Optimization techniques play an important role in solving such problems because they allow systems to evaluate multiple possible scenarios and identify the most beneficial outcome.

Bezditnyi highlights that AI can be used to optimize tax planning by analyzing income, deductions, and tax rules to generate efficient strategies [5]. This shows the potential of

combining machine learning with optimization techniques in taxation systems. However, most existing systems still focus on calculating taxes rather than exploring different strategies and comparing their outcomes.

In practice, taxpayers often rely on human experts such as accountants or auditors to identify tax-saving opportunities. This indicates that current digital systems lack the capability to provide automated and intelligent tax optimization, creating a need for research in this area.

1.2.4 Rule-based systems and tax compliance

Taxation is a rule-based domain, meaning that all calculations and decisions must follow legal regulations defined by authorities such as the Inland Revenue Department [1]. Therefore, any intelligent tax system must include a rule-based component to ensure compliance with tax laws.

Traditional tax systems use rule-based logic to calculate tax amounts based on income categories, deductions, and tax rates. These systems are reliable for ensuring legal correctness but are often limited in flexibility and adaptability. They do not provide intelligent recommendations or analyze different financial scenarios.

Combining rule-based systems with AI techniques can help overcome these limitations. While AI can generate potential strategies, rule-based components can ensure that these strategies remain legally valid. This hybrid approach is important for building reliable tax advisory systems.

1.2.5 Explainable AI in financial systems

Explainability is an important factor in financial and taxation systems because users need to understand how decisions are made. In many AI-based systems, decisions are generated using complex models, which can make them difficult to interpret.

Research in AI emphasizes that transparency is essential for building trust, especially in high-risk domains such as finance and taxation [2]. Users are more likely to trust a system if it clearly explains how a recommendation was generated. In financial advisory systems, explainable AI techniques help users understand why certain decisions or strategies are suggested [6-12].

However, many existing systems still lack clear explanations. They may provide results without showing the reasoning behind them, which can reduce user confidence. This creates a need for

tax advisory systems that not only generate recommendations but also explain them in a simple and understandable way.

1.2.6 Data-driven approaches and challenges in taxation systems

Machine learning and data mining techniques have been widely used in taxation systems for tasks such as fraud detection and risk analysis. Zheng et al. highlight that data-driven approaches can improve tax systems by identifying patterns and detecting inconsistencies in financial data [7]. These techniques help governments and organizations improve compliance and reduce errors.

However, applying data-driven methods to tax planning presents several challenges. Financial data is often sensitive, limited, or difficult to access, which makes it hard to train accurate models. In addition, tax systems vary across countries, making it difficult to develop generalized solutions.

For Sri Lanka, these challenges are even more significant because there are limited publicly available datasets and fewer AI-based tax systems tailored to local regulations. This highlights the need for systems that can work with simulated or structured data while still providing useful recommendations.

1.2.7 Summary of limitations in existing literature

Overall, existing research and systems provide useful insights into digital taxation, AI-based financial advisory, optimization techniques, and rule-based compliance. However, several limitations can be identified. First, most existing systems focus on tax calculation and filing rather than intelligent tax strategy optimization. Second, AI applications in taxation are mainly used for auditing or prediction, not for personalized recommendation. Third, many systems lack explainability, making it difficult for users to trust the results. Fourth, current solutions are not well adapted to Sri Lankan tax regulations and local user needs. Finally, there is a lack of integrated systems that combine optimization, compliance, and explainable decision support into a single platform.

These limitations highlight the need for an intelligent tax strategy optimization system that can analyze financial data, generate optimized tax strategies, ensure legal compliance, and provide clear explanations to users.

1.3 Research Gap

1.3.1 Lack of Intelligent Tax Strategy Optimization Systems

Most existing digital taxation platforms primarily focus on **tax calculation and filing automation** rather than providing intelligent tax strategy recommendations. Commercial tax software systems such as TurboTax and TaxAct assist users in calculating tax liabilities based on predefined rules but do not actively analyze multiple financial scenarios to recommend optimal tax strategies [6]. Similarly, research in taxation systems has largely focused on tax fraud detection or compliance monitoring using machine learning techniques [7].

However, **tax optimization requires analyzing various income categories, deductions, and exemptions simultaneously to identify legally beneficial strategies**. Current systems lack the capability to perform such strategic evaluations, leaving taxpayers dependent on auditors or tax consultants for advisory services. This creates a significant gap in providing **automated and intelligent tax planning assistance**, particularly for individual taxpayers who lack access to professional financial advice.

1.3.2 Limited Adaptation to Country-Specific Tax Regulations

Many existing AI-based financial advisory systems are designed for **general financial planning or global taxation frameworks** rather than country-specific regulatory environments. Tax systems differ significantly across countries due to variations in tax brackets, deductions, relief schemes, and legal requirements.

In Sri Lanka, taxation is governed by the **Inland Revenue Act No. 24 of 2017**, which includes multiple income categories, exemptions, and relief schemes that must be applied correctly to determine tax liability [1]. Despite the availability of online tax filing systems, there is currently **no intelligent platform that analyzes financial data and recommends optimized tax strategies specifically tailored to Sri Lankan tax regulations**.

This lack of localization creates a major limitation for applying existing AI-based tax advisory tools in the Sri Lankan context.

1.3.3 Lack of Explainable AI in Tax Advisory Systems

Trust and transparency are essential in financial decision-making systems. However, many AI-based financial advisory tools operate as **black-box models**, generating recommendations without clearly explaining the reasoning behind them [5]. This lack of explainability can reduce user confidence, particularly in sensitive domains such as taxation where legal compliance is critical.

For taxpayers, understanding **why a particular deduction, exemption, or strategy is recommended** is as important as the recommendation itself. Existing systems rarely integrate **Explainable AI (XAI) techniques** to provide interpretable insights into tax calculations and optimization strategies. As a result, users may hesitate to rely on automated recommendations without clear justification.

1.3.4 Limited Integration of Data Analytics and Decision Support

Although AI and data analytics have been widely applied in financial systems, their use in **integrated tax decision support systems remains limited**. Many current platforms perform isolated tasks such as tax calculation, financial forecasting, or compliance monitoring [6], [7].

However, an effective tax advisory system should combine **financial data processing, tax rule reasoning, optimization algorithms, and recommendation generation within a unified architecture**. The absence of such integration prevents users from receiving **comprehensive tax insights**, forcing them to rely on multiple tools or professional advisors for financial planning.

1.3.5 Need for an Intelligent and Accessible Tax Advisory Platform

Existing tax systems primarily serve as **compliance tools** rather than **decision-support platforms**. As a result, taxpayers often struggle to understand complex tax regulations and identify strategies that minimize tax liability while remaining legally compliant.

This highlights the need for a **data-driven intelligent decision support system** capable of analyzing financial information, applying Sri Lankan tax rules, and generating personalized tax strategy recommendations with clear explanations. Such a system would improve **tax transparency, accessibility, and financial decision-making**, while reducing reliance on manual advisory services.

1.4 Research Problem

Although digital tax systems and financial software are widely available, they do not fully support individuals in making informed tax planning decisions. Most existing platforms, including government e-services and commercial tax software, primarily focus on tax calculation, filing, and basic compliance processes rather than helping users identify optimized tax strategies [1]. While these systems improve efficiency, they still require users to manually understand tax rules, deductions, and financial implications.

In practice, many taxpayers rely on auditors or tax consultants to identify tax-saving opportunities and ensure compliance. This manual approach can be time-consuming, costly, and sometimes inconsistent depending on the experience of the advisor. Even though AI and machine learning have been applied in financial systems, most existing applications in taxation focus on auditing, fraud detection, or prediction rather than personalized tax strategy optimization [3], [7].

Another important limitation is the lack of explainability in many automated systems. Users often receive tax results or recommendations without clear reasoning, making it difficult to understand how different financial decisions affect their tax outcomes. As highlighted in AI research, transparency is essential in decision-support systems, especially in sensitive domains such as taxation [2], [6].

This problem is more significant in the Sri Lankan context, where tax regulations are complex and require careful interpretation of multiple income sources, deductions, and relief schemes defined under the Inland Revenue Act [1]. In addition, there are limited intelligent systems specifically designed for Sri Lankan taxpayers. As a result, many individuals struggle to identify effective tax strategies and may either overpay taxes or fail to fully utilize available benefits.

Therefore, the research problem addressed in this study is the lack of an intelligent, explainable, and user-friendly system that can analyze financial data, generate optimized tax strategies, ensure compliance with Sri Lankan tax regulations, and clearly explain the reasoning behind each recommendation.

1.4.1 System Comparison (Existing Systems vs Proposed System)

Feature	Tax Portals (IRD e-Services)	Accounting Software (QuickBooks, Zoho Books)	Tax Software (TurboTax, etc.)	Existing Research (AI/ML systems)	Proposed System (Our Solution)
Filing support	Yes	Partial	Yes	No	Yes
Tax calculation	Yes	Yes	Yes	Partial	Yes
Tax strategy optimization	No	No	Limited	Limited	Yes
Explainable recommendations	No	No	Limited	Limited	Yes
Personalized tax advice	No	Limited	Limited	Partial	Yes
Compliance with Sri Lankan tax rules	Yes	No	No	No	Yes
Scenario analysis (multiple strategies)	No	No	No	Limited	Yes
Automation level	Medium	Medium	High	High	High
User understanding support	Low	Low	Medium	Medium	High

Table 2:System Comparison

OBJECTIVE

2.1 Main Objective

The main objective of this research is to develop an intelligent, explainable, and data-driven tax strategy optimization system that supports individuals in making effective tax planning decisions while ensuring compliance with Sri Lankan tax regulations.

The proposed system aims to analyze user financial information, including multiple income sources and applicable deductions, and generate optimized tax strategies that can legally minimize tax liability. By integrating machine learning techniques with rule-based tax compliance derived from the Inland Revenue Act, the system will provide recommendations that are both efficient and legally valid.

In addition, the system will incorporate Explainable Artificial Intelligence (XAI) techniques to ensure that all recommendations are transparent and understandable. This will help users clearly understand how different financial decisions influence their tax outcomes.

Ultimately, the research aims to develop a user-friendly decision-support system that enhances tax planning efficiency, reduces dependence on manual advisory services, and improves accessibility, transparency, and trust in digital tax solutions in Sri Lanka.

2.2 Specific Objectives

To achieve the main objective, the following specific objectives are defined:

1. Develop an intelligent tax strategy optimization engine

- Analyze user financial data, including employment, business, and investment income.
- Identify taxable and non-taxable components based on Sri Lankan tax regulations.
- Generate multiple tax strategy options by considering deductions, exemptions, and relief schemes.
- Evaluate and select strategies that minimize tax liability while maintaining compliance.

2. Implement a learning-based optimization mechanism

- Apply machine learning and optimization techniques to analyze financial scenarios.
- Train and test models using simulated taxpayer profiles.
- Improve strategy recommendations through iterative learning and performance evaluation.

3. Design a rule-based tax compliance module

- Encode tax rules from the Inland Revenue Act into a rule engine.
- Validate all generated strategies to ensure legal correctness.
- Prevent invalid or non-compliant tax recommendations.

4. Develop an explainable decision support module

- Apply Explainable AI (XAI) techniques to interpret model outputs.
- Provide clear, human-readable explanations for each recommendation.
- Improve user understanding, transparency, and trust in system decisions.

5. Evaluate system performance and effectiveness

- Compare system-generated strategies with traditional tax planning approaches.
- Measure performance using:
    - tax savings potential
    - compliance accuracy
    - explanation clarity
- Assess system usability across different taxpayer profiles.

METHODOLOGY

This research adopts a structured and systematic methodology to design, develop, and evaluate an intelligent tax strategy optimization and explainable decision support system tailored for Sri Lankan taxpayers. The methodology consists of the research approach, technical methods, tools and platforms, data collection strategy, and evaluation framework used to validate the proposed system.

3.1 Research Approach

This study follows a **Design Science Research (DSR)** approach, which focuses on developing innovative artifacts to solve real-world problems. In this research, the identified problem is the lack of intelligent systems that assist individuals in optimizing tax strategies while ensuring compliance with Sri Lankan tax regulations.

Using this approach, the research designs and develops an **AI-driven decision support system** that analyzes financial data and generates optimized tax strategies. The developed system will be evaluated through experimental analysis using simulated taxpayer scenarios to assess its effectiveness, accuracy, and usability.

3.2 Technical Methods and Algorithms

The proposed system integrates **machine learning, optimization techniques, and rule-based reasoning** to generate intelligent and compliant tax strategy recommendations.

Initially, the system processes user financial data, including income categories, deductions, and applicable tax relief schemes. Based on this input, an optimization mechanism is applied to evaluate multiple tax scenarios and identify strategies that minimize tax liability.

To enhance decision-making, **optimization techniques such as constraint-based optimization and scenario analysis** will be used to explore possible tax strategies efficiently.

A **rule-based compliance layer** is incorporated to ensure that all generated strategies strictly follow Sri Lankan tax regulations defined in the Inland Revenue Act. This layer validates each recommendation and prevents non-compliant outputs.

Furthermore, **Explainable Artificial Intelligence (XAI)** techniques will be integrated to improve transparency. Methods such as SHAP will be used to identify key factors influencing recommendations and generate human-readable explanations for users.

3.3 Tools and Platforms with Justification

The development of the proposed system will utilize a set of tools and technologies selected based on efficiency, scalability, and suitability for AI system development.

Python will be used as the primary programming language due to its extensive support for machine learning and data analysis. Libraries such as **Pandas, NumPy, and Scikit-learn** will be used for data preprocessing, modeling, and analysis.

A **MongoDB database** will be used to store user financial data, tax rules, and generated strategies. Its flexible NoSQL structure allows efficient handling of semi-structured financial data and dynamic tax scenarios.

Development and experimentation will be conducted using **Jupyter Notebook and Visual Studio Code**, which provide an effective environment for building and testing models.

For explainability, **SHAP (SHapley Additive exPlanations)** will be used to interpret model outputs and provide transparent insights into system recommendations.

3.4 Data Collection and Ethical Considerations

Due to the sensitive nature of financial data, this research does not use real personal financial records. Instead, **synthetic datasets and simulated taxpayer profiles** will be generated based on publicly available tax regulations and income structures in Sri Lanka.

The dataset will include variables such as income types, deductions, tax brackets, and relief schemes to simulate realistic financial scenarios. This approach ensures that the system can be trained and evaluated effectively while maintaining user privacy.

All data will be securely stored, and ethical guidelines will be followed to ensure responsible data handling, privacy protection, and compliance with research ethics standards.

3.5 Validation Strategy and Performance Metrics

The proposed system will be evaluated through experimental testing using multiple simulated financial scenarios.

The performance of the system will be compared with traditional rule-based tax calculation approaches to assess improvements in decision-making and optimization.

The following evaluation metrics will be used:

- **Tax savings potential** – measures the reduction in tax liability achieved through optimized strategies
- **Compliance accuracy** – evaluates whether all recommendations adhere to Sri Lankan tax regulations
- **Explainability score** – assesses how clearly the system explains its recommendations to users
- **System performance** – measures execution time and efficiency
- **User interpretability** – evaluates how easily users understand and trust the recommendations

These evaluation criteria ensure that the system is not only accurate and efficient but also transparent and user-friendly.

HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components

The proposed system is an **AI-driven tax strategy optimization and decision support platform** designed for Sri Lankan taxpayers. The system aims to assist users in understanding their tax obligations and identifying legally compliant strategies that minimize tax liability.

The architecture is designed as a **modular and integrated system**, consisting of four main components, each responsible for a specific stage of the tax advisory process.

Component A – Financial Transaction Analysis Module

This module is responsible for processing raw financial transaction data and extracting structured financial information. It classifies transactions into different income categories such as employment income, business income, and investment income using machine learning techniques. The output of this module is a structured financial profile that serves as input for downstream components.

Component B – Tax Strategy Optimization & Explainable Decision Engine

This component serves as the core of the system. It analyzes the structured financial profile and generates optimized tax strategies using learning-based optimization techniques. A rule-based compliance layer ensures that all generated strategies strictly follow Sri Lankan tax regulations.

Additionally, this component integrates Explainable Artificial Intelligence (XAI) methods to generate transparent and human-readable explanations, enabling users to understand how each recommendation contributes to tax reduction.

Component C – Intelligent Tax Advisory Chatbot

This module provides a natural language interface for user interaction. It allows users to query tax-related information, receive explanations, and obtain guidance on tax rules, deductions, and filing procedures. The chatbot enhances system usability and accessibility by providing interactive support.

Component D – Personalized Recommendation and Impact Prediction Engine

This component evaluates multiple tax strategies and predicts their long-term financial impact. It analyzes factors such as projected income growth and user behavior to rank strategies based on expected benefits. This enables users to select the most effective tax planning option.

These components collectively form an **integrated intelligent tax advisory system**, enabling end-to-end analysis, optimization, and explanation of tax strategies.

4.2 High-Level System Architecture Diagram

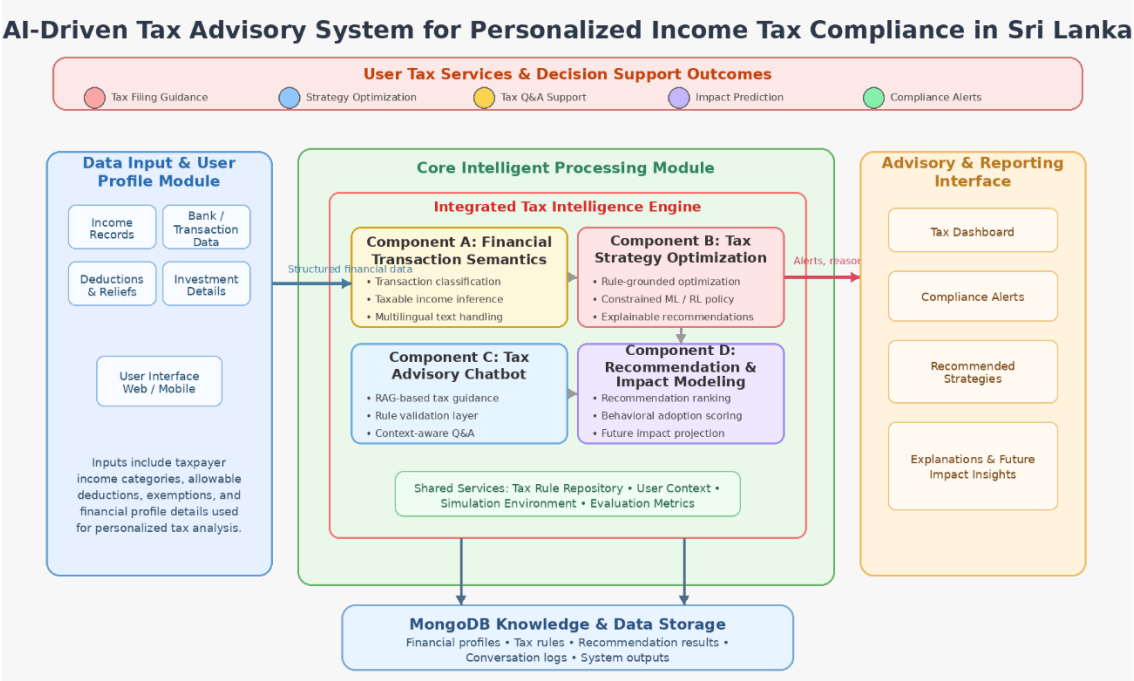


Figure 1:High-Level System Architecture of the AI Tax Advisory System

4.3 Highlight of Individual Component (Component B)

The **Tax Strategy Optimization & Explainable Decision Engine** is the central component responsible for generating optimized and explainable tax strategies.

This module receives structured financial data from Component A and applies optimization algorithms to evaluate multiple tax scenarios. The system identifies strategies that minimize

tax liability while ensuring compliance with Sri Lankan tax regulations through a rule-based validation mechanism.

Furthermore, Explainable AI techniques are integrated to provide transparent reasoning for each recommendation. This allows users to clearly understand how specific deductions, exemptions, or financial decisions influence their tax outcomes.

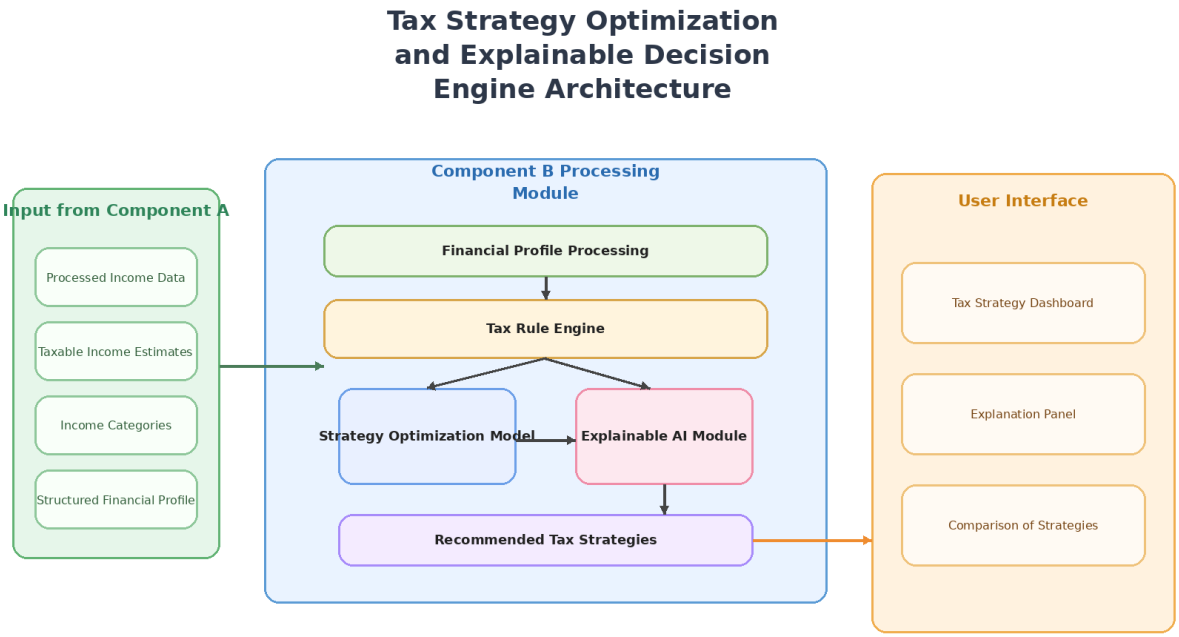


Figure 2: Detailed Architecture of Component B – Tax Strategy Optimization Engine

4.4 Component Interactions

The system components interact through a centralized application layer that facilitates data exchange and coordination.

Initially, financial data is processed by Component A, which generates a structured financial profile. This profile is passed to Component B, where optimized tax strategies are generated. The results are then evaluated by Component D to assess their long-term impact.

Meanwhile, Component C interacts with users by providing explanations and answering tax-related queries. All components communicate with a centralized MongoDB database, which stores financial data, tax rules, generated strategies, and system logs.

4.5 Data Flow Explanation

The system follows a structured data flow process:

1. User inputs financial information through the system interface.
2. Component A processes and categorizes financial transactions.
3. The structured financial profile is forwarded to Component B.
4. Component B generates optimized tax strategies and corresponding explanations.
5. Component D evaluates and ranks strategies based on projected outcomes.
6. Component C provides interactive explanations and user guidance.

The final optimized and explainable tax recommendations are presented to the user through the application interface.

4.6 Scalability and Security Considerations

The system is designed using a **modular and scalable architecture**, allowing each component to operate as an independent service. This enables the system to handle increasing user loads efficiently.

MongoDB is used as the primary database due to its flexibility in handling semi-structured financial data and its scalability for large datasets.

From a security perspective, sensitive financial data will be protected through encryption during both storage and transmission. Authentication and access control mechanisms will be implemented to ensure that only authorized users can access personal data.

Additionally, the system will follow best practices in **data privacy, security, and ethical data handling**, ensuring compliance with modern data protection standards.

USER REQUIREMENTS

This section defines the key stakeholders and system requirements of the proposed AI-driven tax strategy optimization and decision support system. The system is designed to assist Sri Lankan taxpayers in understanding tax obligations and identifying legally compliant tax-saving strategies through an intelligent recommendation framework.

5.1 Stakeholder Identification

The proposed system involves multiple stakeholders who interact with the platform either directly or indirectly.

Individual Taxpayers

Individual taxpayers represent the primary users of the system. These users are responsible for submitting annual income tax returns and managing their personal financial data. The system supports them by analyzing their financial information and providing optimized tax strategy recommendations that comply with Sri Lankan tax regulations.

Tax Consultants and Auditors

Tax professionals such as auditors and consultants can use the system as a decision-support tool. It assists them in analyzing financial scenarios, identifying tax-saving opportunities, and providing advisory services more efficiently through automated recommendations and explainable outputs.

Inland Revenue Department (IRD)

The Inland Revenue Department acts as a regulatory stakeholder. Although IRD officials may not directly interact with the system, their tax policies and regulations form the foundation of the rule-based compliance layer. Therefore, the system must strictly adhere to the Inland Revenue Act and related guidelines.

System Administrators

System administrators are responsible for managing the technical infrastructure of the platform. Their responsibilities include database management, system maintenance, updates, and ensuring system security and availability.

5.2 Functional Requirements

The functional requirements define the core system functionalities required to achieve the objectives of the proposed system.

FR1 – Financial Data Input

The system shall allow users to input financial information, including income sources, deductions, and investment details.

FR2 – Financial Profile Analysis

The system shall process and analyze user financial data to classify income according to Sri Lankan tax categories.

FR3 – Tax Rule Application

The system shall apply tax rules defined in the Inland Revenue Act to validate calculations and ensure compliance.

FR4 – Tax Strategy Optimization

The system shall generate optimized tax strategies that minimize tax liability while maintaining legal compliance.

FR5 – Recommendation Explanation

The system shall provide clear, explainable insights showing how each recommendation impacts the user's tax outcome.

FR6 – Strategy Comparison

The system shall enable users to compare multiple tax strategies and evaluate their financial benefits.

FR7 – Data Storage and Retrieval

The system shall securely store and retrieve user data, tax rules, and generated outputs using a database system such as MongoDB.

5.3 Non-Functional Requirements

The non-functional requirements define the quality attributes and operational constraints of the system.

Security

The system shall ensure data security through encryption, secure authentication, and access control mechanisms to protect sensitive financial information.

Performance

The system shall process user data and generate tax recommendations within acceptable response times.

Usability

The system interface shall be user-friendly and easy to navigate, allowing individuals with limited technical knowledge to use the platform effectively.

Scalability

The system shall be designed to handle an increasing number of users and data without performance degradation.

Reliability

The system shall produce consistent and accurate results based on updated Sri Lankan tax rules.

Data Privacy

The system shall ensure responsible handling of user data and comply with ethical data usage and privacy standards.

COMMERCIALIZATION PLAN

The proposed **AI-Driven Tax Strategy Optimization and Explainable Decision Support System** is designed to address the growing need for intelligent and accessible tax advisory solutions in Sri Lanka. Many taxpayers face difficulties in understanding complex tax regulations and often rely on manual consultation services, which can be time-consuming and costly.

The system provides a **digital, AI-powered platform** that enables users to analyze financial data, generate optimized tax strategies, and understand tax obligations through transparent and explainable recommendations. By integrating machine learning, optimization techniques, and rule-based compliance aligned with the Sri Lankan Inland Revenue Act, the platform offers a scalable and practical solution for modern tax planning.

The commercialization strategy focuses on delivering the system as a **Software-as-a-Service (SaaS) platform**, targeting both individual users and professional financial service providers.

6.1 Target Market and Customer Persona

The primary target market consists of **individual taxpayers in Sri Lanka**, including salaried employees, freelancers, and small business owners who are required to submit income tax returns.

Many individuals in this segment have limited understanding of tax regulations and depend on external advisors to identify deductions, exemptions, and tax-saving opportunities. The proposed system provides these users with **personalized, data-driven tax strategies**, reducing reliance on manual advisory services.

A secondary market includes **tax consultants, auditors, and accounting firms**, who can use the system as a decision-support tool to improve efficiency and accuracy in client advisory services.

Additionally, **small and medium-sized enterprises (SMEs)** can benefit from the platform by using it to evaluate tax planning strategies and support financial decision-making.

6.2 Value Proposition

The key value of the proposed system lies in its ability to deliver **intelligent, explainable, and personalized tax advisory services**.

Unlike traditional tax software that focuses primarily on tax calculation and filing, the proposed system:

- Generates **optimized tax strategies** based on user financial data
- Ensures **legal compliance** through rule-based validation
- Provides **explainable insights** to improve user understanding

This combination of optimization, compliance, and explainability enables users to make informed financial decisions while improving trust in automated tax systems.

Furthermore, the platform reduces dependency on costly professional consultations and offers a more **accessible and user-friendly tax planning solution**.

6.3 Revenue Model

The proposed system adopts a **freemium-based SaaS revenue model**.

Basic features such as general tax guidance and simple calculations will be provided free of charge to attract a broad user base. Advanced features, including:

- personalized tax strategy optimization
- scenario-based financial analysis
- detailed explainable recommendations

will be offered through **premium subscription plans**.

Additional revenue streams may include:

- **Professional subscriptions** for auditors and accounting firms
- **Enterprise solutions** for financial service providers
- Potential partnerships with fintech platforms

6.4 Cost Estimation

The major cost components associated with system development and operation include:

- **Development Costs**: Implementation of machine learning models, optimization algorithms, rule-based systems, and user interface design
- **Infrastructure Costs**: Cloud-based hosting, database management, and system scalability
- **Operational Costs**: System maintenance, updates, security management, and customer support

Cloud infrastructure will play a critical role in ensuring system availability, scalability, and secure storage of financial data.

6.5 Pricing Strategy

The system will adopt a **tiered pricing strategy** suitable for the Sri Lankan market.

- **Free Tier**: Basic tax calculation and general guidance
- **Standard Tier**: Access to tax optimization features and limited scenario analysis
- **Premium Tier**: Full access to advanced features, including explainable recommendations and financial impact analysis

For professional users, **enterprise-level pricing plans** will provide additional analytical tools and multi-client support capabilities.

6.6 Competitive Advantage

The proposed system offers several competitive advantages compared to existing tax software and advisory services:

- **AI-driven optimization** for generating effective tax strategies
- **Explainable AI integration**, improving transparency and user trust
- **Compliance-focused design** aligned with Sri Lankan tax regulations
- **Scenario-based analysis**, enabling comparison of multiple tax strategies

These features position the system as a **next-generation tax advisory platform**, going beyond traditional tax calculation tools.

6.7 Intellectual Property Considerations

The proposed system includes innovative components such as:

- the **tax strategy optimization engine**
- the **explainable decision support framework**

These components can be protected through **software copyrights and intellectual property registration**.

Protecting these innovations will support commercialization efforts, maintain competitive advantage, and ensure that the core system design and algorithms remain proprietary.

BUDGET AND JUSTIFICATION

The development of the proposed **AI-Driven Tax Strategy Optimization and Explainable Decision Support System** requires a combination of hardware resources, software tools, cloud infrastructure, and human effort. The estimated budget is designed to support all stages of system development, including implementation, testing, and deployment.

7.1 Development Hardware

The system development requires a reliable computing device such as a laptop or workstation capable of handling machine learning model development, data processing, and system integration tasks.

This hardware will be used for:

- training and testing machine learning models
- running optimization algorithms
- debugging and system integration
- performance evaluation

Adequate computational resources are essential to ensure efficient development and smooth execution of system components.

7.2 Software Licenses

The majority of tools used in this project are **open-source technologies**, including Python and its associated machine learning libraries, which significantly reduce development costs.

However, certain tools may require licensed versions for:

- user interface design
- documentation and report preparation
- advanced development features

These tools improve productivity and support the development of a professional-quality system.

7.3 Cloud Services

Cloud infrastructure will be used to host the application and manage data storage securely. This enables the system to provide reliable access to users and ensures high availability.

Cloud services support:

- secure storage of financial data
- scalability to handle multiple users
- system performance and uptime

This is essential for deploying the system as a scalable and accessible digital platform.

7.4 Human Effort Cost

Human effort represents a significant component of the development process. The project requires technical expertise in system design, machine learning, optimization, and software development.

Key activities include:

- system architecture design
- implementation of the tax rule engine
- development of machine learning and optimization models
- integration of explainable AI components
- system testing and performance improvement
- preparation of research documentation

This effort ensures the successful design and evaluation of the proposed system.

7.5 Justification for Each Expense

Each budget component is essential for the successful development and deployment of the system.

- **Hardware resources** provide the computational power required for development and testing
- **Software tools** support efficient system design and implementation
- **Cloud services** ensure scalability, accessibility, and secure data management
- **Human effort** represents the time and expertise required to develop and evaluate the system

Together, these resources ensure that the system can be developed effectively while maintaining quality, performance, and reliability.

Estimated Budget

Resource Type	Amount (LKR)
Development Hardware (Laptop / Workstation usage)	150,000
Software Tools and Development Utilities	20,000
Cloud Hosting Services	50,000
Data Processing and Testing Resources	20,000
Miscellaneous Development Expenses	10,000
Total Estimated Budget	**250,000**

Table 3:Estimated Budget

WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) represents the hierarchical decomposition of the tasks involved in developing the proposed **AI-Driven Tax Strategy Optimization and Explainable Decision Support System**. It organizes the overall project into well-defined phases to ensure systematic planning, development, and evaluation.

The project is divided into several major phases, including:

- **Research and Problem Analysis**
- **System Design and Planning**
- **Data Preparation and Processing**
- **Model Development and Optimization**
- **System Implementation and Integration**
- **Testing and Evaluation**
- **Documentation and Finalization**

Each phase consists of a set of specific tasks required to achieve the project objectives.

The initial phases focus on understanding the problem domain through literature review, identifying research gaps, and designing the system architecture. These stages establish the foundation for the development of the proposed solution.

The development phases involve implementing machine learning models, designing the tax strategy optimization engine, and integrating the rule-based compliance layer. These components are combined to form the core functionality of the system.

The final phases include system testing, performance evaluation, and validation using simulated financial scenarios. This ensures that the system produces accurate, reliable, and explainable tax recommendations.

By organizing the project into structured phases, the WBS provides a clear roadmap for development, facilitates progress tracking, and ensures that each stage of the system is completed efficiently within the defined project timeline.

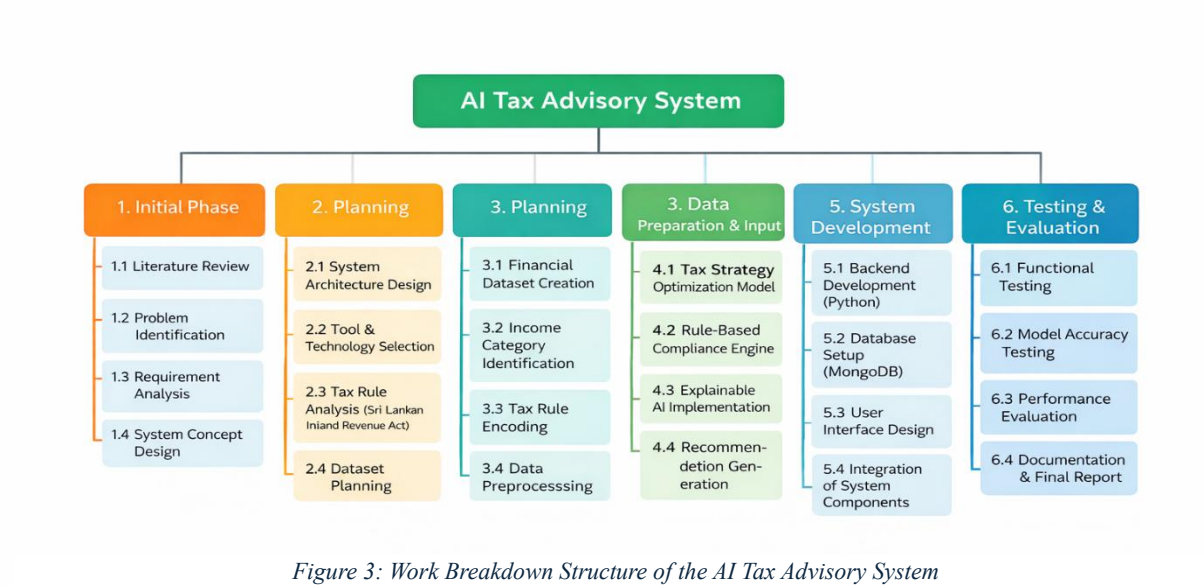


Figure 3: Work Breakdown Structure of the AI Tax Advisory System

8.1 Task Breakdown per Member

In this project, each team member is responsible for developing one key component of the system. Their tasks include system design, implementation of models or algorithms, testing the component, and supporting documentation.

Team Member	Component Responsibility	Key Tasks
Member 1	Financial Transaction Analysis	Data preprocessing, transaction classification, and feature extraction from financial records
Member 2	**Tax Strategy Optimization & Explainable Decision Engine**	Development of tax optimization models, implementation of Sri Lankan tax rules, and integration of explainable AI techniques
Member 3	Tax Advisory Chatbot	Development of chatbot interaction, natural language processing, and user query handling
Member 4	Recommendation & Impact Prediction	Strategy recommendation ranking, prediction models, and financial impact analysis

Table 4:Task Breakdown Per Member

8.2 Timeline Alignment

The project will be carried out in several development phases to ensure that the system is completed within the planned time period. These phases include research, system development, testing, and final evaluation.

Each phase focuses on specific tasks that contribute to building the final system.

Phase	Activities	Duration
Phase 1	Literature review and system design	Month 1 – Month 2
Phase 2	Data preparation and tax rule engine development	Month 3 – Month 4
Phase 3	Machine learning model development	Month 5 – Month 7
Phase 4	System integration and testing	Month 8 – Month 9
Phase 5	Evaluation, improvements, and documentation	Month 10 – Month 12

Table 5:Project Development Timeline

8.3 Balanced Workload Distribution

The project tasks are carefully distributed among the team members so that each person contributes equally to the development process. Since each component represents an important part of the system, team members can work independently on their assigned modules while collaborating during the system integration phase.

Regular team meetings and progress discussions will be conducted to review the development status, address technical challenges, and ensure that the project remains aligned with the proposed timeline. This collaborative approach helps maintain steady progress and ensures the successful completion of the **AI Tax Advisory System designed for Sri Lankan users**.

GANTT CHART

The Gantt chart illustrates the planned timeline for the development of the AI Tax Advisory System over a 12-month period. It outlines the major phases of the project, including requirements analysis, data preparation, model development, system integration, testing, and final deployment. This timeline helps ensure that each stage of the project is completed in an organized manner while maintaining steady progress toward the final system implementation.

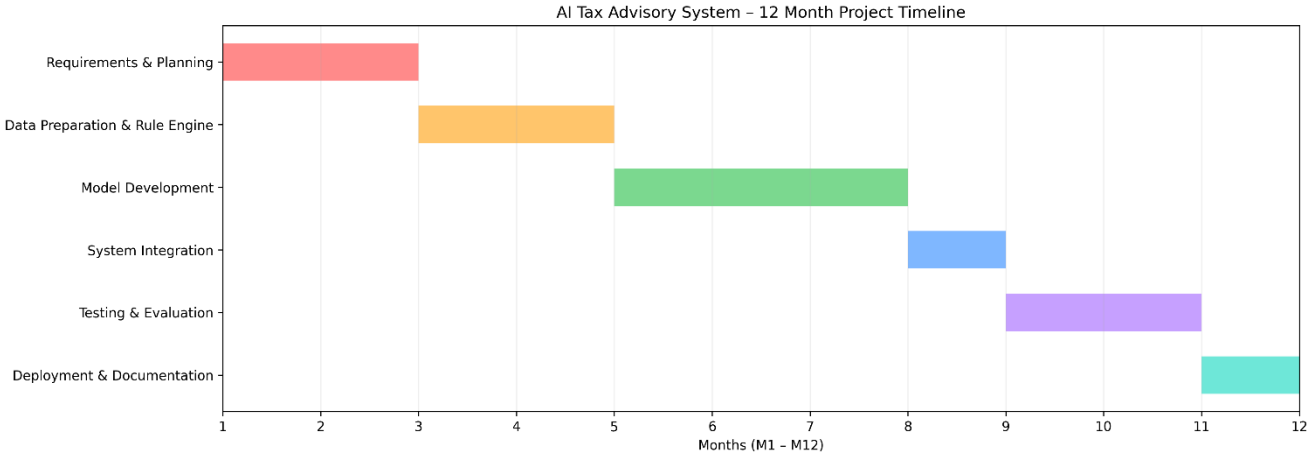


Figure 4: Project Timeline Gantt Chart

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LIST OF APPENDICES

Appendix A - High-Level Architecture of the Tax Strategy Optimization & Explainable Decision Engine (Component B)
Appendix B - Work Breakdown Structure (WBS) Diagram of the AI Tax Advisory System
Appendix C - Project Timeline Gantt Chart